ClipSAT : Contest Math & Olympiad AMC 10/12, Kangaroo & Mawhiba [ANSWER KEY]

Unit 5.01: Recurrence Relations, Markov Chains & Geometric Probability

Q1. The coordinate diagram shows the parabolic curve f_1x = x^2 - 14x + 64 in the domain of Recurrence Relations & Geometric Probability. By completing the square f_1x = x - h^2 + k , find the sum k + f_19 .

[x] A: 34

[ ] B: 41

[ ] C: 19

[ ] D: 32

Solution: **Mark Scheme & Examiner Solution Recurrence Relations & Geometric Probability:** 1. Complete the square: x^2 - 14x + 64 = x - 7^2 - 49 + 64 = x - 7^2 + 15 , giving minimum value k = 15 . **[M1]** 2. Evaluate at x = 9 : f_19 = 2^2 + 15 = 19 . **[M1]** 3. Sum the two quantities: k + f_19 = 15 + 19 = 34 . **[A1]** - **Final Verified Answer:** 34 .

Q2. A mathematical model in Recurrence Relations & Geometric Probability is defined by the cubic function y_2x = 3x^3 - 7x^2 + 15x - 12 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[ ] A: 96

[x] B: 54

[ ] C: 39

[ ] D: 28

Solution: **Mark Scheme & Examiner Solution Recurrence Relations & Geometric Probability:** 1. Differentiate term by term using the power rule: frac{mathrm{d}y}{mathrm{d}x} = 9x^2 - 14x + 15 . **[M1]** 2. Substitute x = 3 : left.frac{mathrm{d}y}{mathrm{d}x}right|_{x=3} = 99 - 143 + 15 = 54 . **[M1][A1]** - **Final Verified Answer:** 54 .

Q3. In Recurrence Relations & Geometric Probability, two displacement vectors are given by mathbf{u}_{3} = langle 7, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[ ] A: 7

[ ] B: 113

[x] C: 120

[ ] D: 232

Solution: **Mark Scheme & Examiner Solution Recurrence Relations & Geometric Probability:** 1. Compute the inner product: mathbf{u}_{3} cdot mathbf{v}_{3} = 79 + -87 = 63 + -56 = 7 . **[M1]** 2. Compute the squared norm: |mathbf{u}_{3}|^2 = 7^2 + -8^2 = 113 . **[M1]** 3. Combine: 7 + 113 = 120 . **[A1]** - **Final Verified Answer:** 120 .

Q4. The linear growth profile shown in the diagram for Recurrence Relations & Geometric Probability follows an arithmetic progression with first term a_1 = 8 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[ ] A: 92

[ ] B: 840

[ ] C: 1380

[x] D: 750

Solution: **Mark Scheme & Examiner Solution Recurrence Relations & Geometric Probability:** 1. Use the sum formula for an arithmetic series: S_n = frac{n}{2}[2a_1 + n - 1d] . **[M1]** 2. Substitute n = 15, a_1 = 8, d = 6 : S_{15} = frac{15}{2}[16 + 14 times 6] = frac{15}{2} times 100 = 750 . **[M1][A1]** - **Final Verified Answer:** 750 .

Unit 5.01: Recurrence Relations, Markov Chains & Geometric Probability

Q5. In Recurrence Relations & Geometric Probability, evaluate the exact definite integral I_5 = int_0^{2} 18x^2 + 28x,mathrm{d}x .

[x] A: 104
[ ] B: 128
[ ] C: 132
[ ] D: 53

Solution: **Mark Scheme & Examiner Solution Recurrence Relations & Geometric Probability:** 1. Integrate term by term: int 18x^2 + 28x,mathrm{d}x = left[6x^3 + 14x^2right]_0^{2} . **[M1]** 2. Substitute the upper and lower limits: 62^3 + 142^2 - 0 = 48 + 56 = 104 . **[M1][A1]** - **Final Verified Answer:** 104 .

Q6. In Recurrence Relations & Geometric Probability, consider the quadratic function f_6x = x^2 - 16x + 151 . Express f_6x in the completed-square form x - h^2 + k and compute the exact value of k + f_615 .

[ ] A: 231
[x] B: 223
[ ] C: 136
[ ] D: 216

Solution: **Mark Scheme & Examiner Solution Recurrence Relations & Geometric Probability:** 1. Complete the square: x^2 - 16x + 151 = x - 8^2 - 64 + 151 = x - 8^2 + 87 , giving minimum value k = 87 . **[M1]** 2. Evaluate at x = 15 : f_615 = 7^2 + 87 = 136 . **[M1]** 3. Sum the two quantities: k + f_615 = 87 + 136 = 223 . **[A1]** - **Final Verified Answer:** 223 .

Q7. The graph illustrates the cubic model y_7x = 3x^3 - 8x^2 + 33x - 19 studied in Recurrence Relations & Geometric Probability. Find the instantaneous gradient frac{mathrm{d}y}{mathrm{d}x} of the tangent to the curve at x = 8 .

[ ] A: 609
[ ] B: 448
[x] C: 481
[ ] D: 89

Solution: **Mark Scheme & Examiner Solution Recurrence Relations & Geometric Probability:** 1. Differentiate term by term using the power rule: frac{mathrm{d}y}{mathrm{d}x} = 9x^2 - 16x + 33 . **[M1]** 2. Substitute x = 8 : left.frac{mathrm{d}y}{mathrm{d}x}right|_{x=8} = 964 - 168 + 33 = 481 . **[M1][A1]** - **Final Verified Answer:** 481 .

Q8. In Recurrence Relations & Geometric Probability, two displacement vectors are given by mathbf{u}_{8} = langle 8, -7 rangle and mathbf{v}_{8} = langle 8, 5 rangle . Compute the exact scalar value of mathbf{u}_{8} cdot mathbf{v}_{8} + |mathbf{u}_{8}|^2 .

[ ] A: 29
[ ] B: 113
[ ] C: 212
[x] D: 142

Solution: **Mark Scheme & Examiner Solution Recurrence Relations & Geometric Probability:** 1. Compute the inner product: mathbf{u}_{8} cdot mathbf{v}_{8} = 88 + -75 = 64 + -35 = 29 . **[M1]** 2. Compute the squared norm: |mathbf{u}_{8}|^2 = 8^2 + -7^2 = 113 . **[M1]** 3. Combine: 29 + 113 = 142 . **[A1]** - **Final Verified Answer:** 142 .